ANGELHACK · BUILDING WITH IBM BOB · TRACK 1: EXPLORE, FIX, BUILD

Prove every claim.

Or it doesn’t ship.

I gave IBM Bob a real open-source app students use and one rule. Bob built its own toolkit to enforce it, shipped real users’ requests, and caught itself every time it was wrong.

David J · solo · early career

IBM Bob Shell · 22 runs + Bob in CI

160 tests · axe 18 → 0 · CI green

TOMODORO TUNE-UP

built with IBM Bob Shell

upstream issues shipped	3
upstream bug fixed	1
unit + boot + kit tests	160
axe violations	18 → 0
Bob mistakes caught	8
custom modes · hooks	3 · 2
every prompt + cost	LOGGED
every claim	CHECKED
every push	CI GREEN

keep this receipt

THE PROBLEM

AI writes real code now. It sounds just as sure when it’s wrong.

seen in round 1

Confident, wrong

Bob’s first map of the app listed suspect areas with line numbers. When I read the cited lines, 2 of the claims were false.

seen in round 2

Green tests, broken app

A Bob change passed every unit test, yet the app crashed on load. Unit tests alone would have shipped it.

seen in round 2

“Done” isn’t done

Bob reported spoken reminders done. The code would have spoken to users who never turned the feature on.

So the question isn’t “can Bob write it?” It’s “can Bob prove it?”

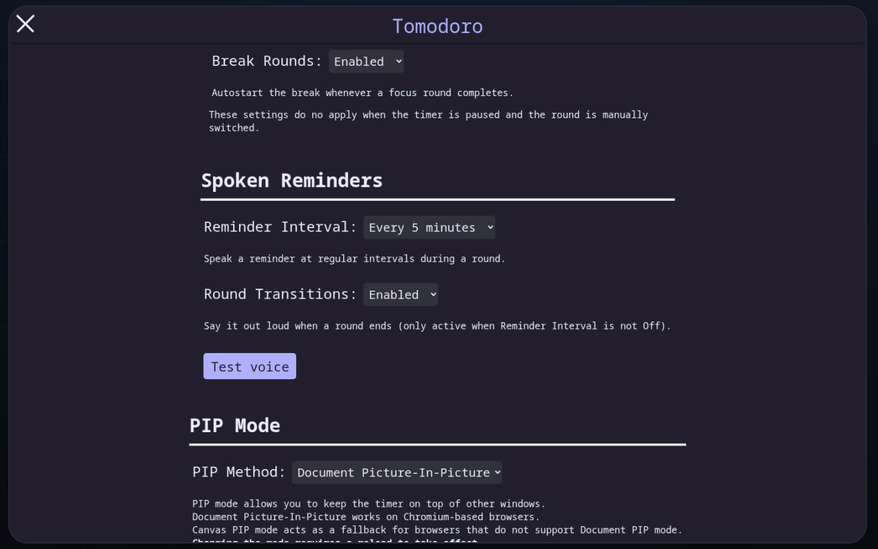
THE PROJECT · EXPLORE → FIX → BUILD

lazy-guy/tomodoro

★ 308 | 55 forks | vanilla JS · PWA | 0 tests at start

A real pomodoro timer students use, with feature requests left open for years. Not a toy built for the demo: a codebase I had never seen, with real users.

- **Explore:** Bob maps the code, every claim cited to file:line
- **Fix:** failing test first, then the fix, with tests rerun automatically
- **Build:** ship the open user requests, then audit Bob’s own work



The tuned app: spoken reminders settings (issue #11)

DEPTH OF BOB ADOPTION · THE TUNE-UP KIT

First, Bob built its own toolkit.

.bob/ · authored by Bob, run 8

```
.bob/
├─ custom_modes.yaml
│   ├─ scout            read-only
│   ├─ test-first-fixer
│   └─ claim-checker
├─ skills/issue-to-fix/
├─ rules/house-rules.md
└─ hooks/
    ├─ test-after-write  PostToolUse
    └─ shell-guard       PreToolUse*
```

S **Scout**

Maps the code and must cite the line for every claim.

T **Test-First Fixer**

Has to show a failing test before it may change code.

C **Claim Checker**

Audits Bob’s own summaries against the code and the prompt.

H **The hook is the receipt**

After every file Bob writes, the full test suite runs and the score goes back to Bob.

* shell-guard written by hand after Bob Shell ran out of memory on chained piped commands (documented in BOB-SHELL-NOTES.md). Plan mode was used for the round-2 plan.

THE MOMENT · BOB CATCHING BOB

Bob said “done.” Its own Claim Checker said no.

run 11 · Test-First Fixer

```
▸ issue #11 · reminders
+ src/speech.js  27 tests
+ settings panel
+ round-end voice
```

✓ DONE

RUN 16 · CLAIM CHECKER

```
 76  let speechTransitions = true;
188  if (speechTransitions) speak(…)
```

F1 new users hear speech they never turned on
F2 speech on a manual skip while paused
F3 an edited note saved twice

RUN 17 · FIXED, TEST FIRST

```
shouldAnnounceTransition()
createEditSession()
+14 tests
```

Speech now only when reminders are on and a round really ended. Verified again in the browser (flows 23–25).

The kit caught what Bob’s own summary missed, before any user heard it.

Every step is a file: docs/bob-runs/11-*, docs/reviews/round2-check.md, commit 911ff48.

WHAT SHIPPED FOR STUDENTS

Real requests from real users, shipped.

#11 shipped

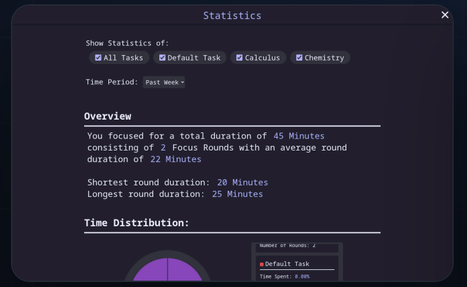
Spoken reminders

For a student with ADHD: time left every 1/5/10/30 min, round-end voice. Silent until turned on.

#8 shipped

Notes on focus rounds

Saved with the round, editable in records, kept in backups.





upstream bug fixed

Picker vs. logged task

After a reload the picker showed one task while time went to another. Also in the original app.

accessibility

axe-core 18 → 0

Labels, landmarks, live round announcements, Escape closes panels, visible focus.

#6 shipped · round 1

A colour per subject

Distinct task colours in the stats pie and bars.

#13 checked

Already solved

Bob's plan showed autostart exists upstream. Documented, not claimed.

THE RECEIPTS

Measured, not claimed.

160
unit + boot + kit tests
(upstream had 0)

35/35
browser flows, desktop
and a 390px phone

18 → 0
axe-core violations in
4 app states

8
times Bob was wrong
and it got caught + fixed

What Bob got wrong	What caught it
Hook always reported 0 tests	running the hook → Bob fixed it test-first (run 9)
App crashed on load, all unit tests green	headless browser check → boot test (run 13)
Speech without opt-in · on skip · note saved twice	Claim Checker (run 16) → fixed (run 17)
Space in the note field started the timer	recording the demo → fixed (run 18)
CI report split Bob’s answer into fragments · showed Bobcoins as $	running it on a real Bob stream → fixed test-first (run 22)

22 Bob runs + 1 Claim Checker run in CI, ≈36 of 50 trial Bobcoins. Every prompt, output, cost and correction is in BOB-LOG.md and docs/bob-runs/.

HOW IT WORKS

Where Bob sits in the pipeline.

1 · EXPLORE

Scout + Plan mode

Architecture map and touch points, every claim with file:line.

→

2 · FIX

Test-First Fixer

Red test, fix, green. The hook reruns the suite after each write.

→

3 · BUILD

issue-to-fix skill

GitHub issue → plan → tests → feature → docs.

→

4 · VERIFY

Claim Checker + me

Audit of Bob's claims, browser flows, axe-core, then a human read.

how each run was started (headless, logged)

```
bob run -f stream-json --mode test-first-fixer --max-cost 3 < docs/bob-runs/NN-task.prompt.txt
```

Tech: IBM Bob Shell 2.0.6 · vanilla JS/HTML/CSS · service worker · IndexedDB · Web Speech API · Node 24 node:test · Playwright · axe-core.

ROUND 3 · OCT 18

The receipts run themselves.

Don’t trust our test report: every push re-runs it on GitHub’s machines, and Bob’s own Claim Checker audits the newest commit in CI.

CI · EVERY PUSH

160 tests · 35/35 flows · axe 0

Unit, boot and kit tests, then the browser flows and axe-core in Chromium. Bob (run 20) made the harness exit 1 on any failure; before, it could never go red.

BOB IN CI

Claim Checker on every PR

Bob Shell installs headless, runs the claim-checker mode on the newest commit (cap 1.5 Bobcoins) and writes its verdicts to the job summary. Written by Bob (runs 21–22).

KIT SELF-TEST

The hooks, tested as Bob calls them

JSON events on stdin: the shell guard blocks the command shape that crashed Bob Shell; the test hook reports “2 passed, 1 failed” on a sample project.

first Claim Checker run in CI · audit of commit fd02d74 · 0.23 Bobcoins · 12 tool calls

```
CONFIRMED     22   e.g. verdict.cjs:43  return parts.join("");
WRONG          0
UNVERIFIABLE   1   "test written first" can't be proven from one commit; it's in the run log
```

What I learned

- Read the line Bob cites before acting on the claim. That habit caught the most mistakes.
- Green unit tests aren’t a working app. A boot test closed that gap for good.
- Modes that force evidence beat longer prompts. Rules Bob can’t skip do the checking.
- Small budgets per run (--max-cost) kept every change reviewable.

Next

- Offer #11, #8 and the upstream bug fix back to lazy-guy/tomodoro as pull requests.
- Publish the Tune-Up Kit as a drop-in .bob/ template for any repo.
- **Done Oct 18:** Claim Checker runs in CI. Next: let a WRONG verdict block the pull request (today it informs, it does not block).
- Next round: headed-browser tests for Picture-in-Picture and notifications, the two features not covered yet.

TEAM · LINKS

Explore. Fix. Build.
With receipts.

TEAM

David J, solo · student / early career

TRY IT

tomodoro-bob.sites.tab.bot/tuneup.html
davidjsourcing.github.io/tomodoro-tune-up

CODE (MIT)

github.com/davidjsourcing/tomodoro-tune-up
gitlab.com/davidj-builds/tomodoro-tune-up

DISCLOSURE

Code changes by IBM Bob Shell, driven and checked by me. An AI assistant helped write prompts, check Bob's claims, and make the video and deck. The narration is text-to-speech.